

ADNAN MASHRUR SADAD

APPLIED AI ENGINEER | LLM SYSTEMS AND MLOPS

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Universiti Teknologi Malaysia (UTM), MJIIT

Kuala Lumpur, Malaysia

B.Sc. Computer Science (Software Engineering) with Honours — CGPA 3.43/4.00

2021 – 2026

- Dean's List 2022/2023 and 2025/2026.

TECHNICAL SKILLS

GenAI / LLM: RAG, agentic workflows and tool calling, Text2SQL, structured-output validation and repair loops, LLM evaluation and benchmarking, prompt engineering, Groq API, Hugging Face, AWQ/NF4 int4 quantization

ML & Data: Machine Learning, Deep Learning, NLP, XGBoost, scikit-learn, PyTorch, TensorFlow, pandas, NumPy, drift detection (PSI/KS), experimental design

Languages: Python, SQL, Java, TypeScript, JavaScript, C++

Backend & Infra: FastAPI, Pydantic, SQLAlchemy, REST APIs, Docker, Kubernetes, Kafka/Redpanda, PyFlink, Feast, MLflow, ONNX, CI/CD (GitHub Actions), Git, Agile

RESEARCH

Deceptive Grounding in Regulatory RAG (SAFE-RAG)

2025 – 2026

First author — short paper under review at ALTA 2026, supervised at UTM MJIIT | [GitHub Link](#)

- Measured an 8.6% base rate (95% CI 4.5–13.5%) of answers that pass schema validation and are entailed by the passage they cite yet remain wrong, because that passage governs a different regulated entity — across 200 items annotated from 2,000 pipeline outputs over the ObliQA financial-regulation corpus.
- Ran the study pre-registered, reaching Cohen's $\kappa = 0.786$ (27/30 exact agreement) after one registered annotation-guideline revision round.

Answerability under Schema Evolution in Text-to-SQL

2026

First author — full manuscript complete (v1.0); not yet submitted

- Built a 1,382-instance benchmark over 20 Spider databases and seven schema-drift families, separating drift a model can observe from drift that is structurally invisible to it.
- Ran 27,384 generations across 39 experimental cells on free-tier Kaggle T4 GPUs, isolating where prompting interventions fail to help at all.
- Shipped a detector recovering 79.5% of the structurally invisible region at a 13.0% false-alarm rate, or 51.2% at zero false alarms, evaluated against 5,627 benign-update variants with labels double-annotated to $\kappa = 0.918$.

PROJECTS

Driftline — Streaming Fraud Detection and Drift-Aware ML Platform

2026

Python, XGBoost, Redpanda/Kafka, PyFlink, Feast, ONNX, FastAPI, Kubernetes, MLflow | [GitHub Link](#)

- Kept a decaying fraud model in service automatically by building PSI-triggered retraining behind a shadow promotion gate: candidate PR-AUC 0.4802 against incumbent 0.2811 on the same holdout, with both triggered retrains passing the gate.
- Replayed 590K IEEE-CIS transactions through Redpanda and PyFlink into Redis and Parquet feature stores, then traced an 8.66% online/offline skew to sink buffering.
- Served ordinal-encoded XGBoost via ONNX and FastAPI at 380 ms p95 under 50 concurrent users on a 4-vCPU GCP VM, handling 10,018 requests with zero failures across five passing CI jobs.

InterviewPilot — AI Mock Interview Platform

2026

FastAPI, SQLAlchemy, React + TypeScript, Groq (Whisper-large-v3, Llama 3.3 70B), Docker, GitHub Actions | [GitHub Link](#)

- Built a full-stack mock-interview platform using Whisper-large-v3 for transcription and Llama 3.3 70B for rubric-based answer evaluation, with persistent resumable sessions and shareable reports.
- Made model output safe to consume downstream by returning schema-validated JSON through a repair loop that re-prompts the model on its own invalid output, plus an agentic follow-up generator that decides when to probe deeper.
- Shipped behind 89% statement coverage across 19 backend tests with Groq calls mocked, Dockerized deployment and GitHub Actions CI on every push.

Agentic Planner — Validated Tool Execution Layer

2026

Python, typed validation, provenance tracking, deterministic testing | [GitHub Link](#)

- Closed a class of silent-failure bugs in agent tool use by enforcing argument provenance, dependency checks and confirmation gates, and by rejecting repairs that relabel unsupported values to bypass validation.
- Removed ambiguous monetary conversion with exact decimal-to-minor-unit validation, verified by 84 tests and four end-to-end scenarios on deterministic mock tools.

EXPERIENCE

Software Developer Intern (Hybrid)

Oct 2024 – Mar 2025

Joget Inc. — enterprise workflow automation on the Joget DX low-code platform

Kuala Lumpur, Malaysia

- Shipped workflow automations and UI components used by client teams across [N] departments, owning the full SDLC in Agile sprints alongside senior engineers, QA and product managers.
- Removed manual re-entry steps from recurring client workflows by integrating REST APIs with external services.
- Maintained and released modules through CI/CD across [N] recurring production release cycles for enterprise clients.